

Filter-feeders

Instead of hunting animals one by one, filter-feeders strain food from the water around them. This way of feeding is very efficient and is used by many animals in freshwater and the oceans. The biggest by far is the blue whale, which feeds mainly on finger-sized crustaceans called krill. Filter-feeders also include birds, fish, and worms, as well as huge numbers of other invertebrates, from mussels and barnacles on rocky shores to jellylike salps in the seas. Most filter-feeders eat live animals and microorganisms, but some behave more like scavengers. They sieve particles of dead matter that drift with the currents or that fall down from the surface waters above.

Record appetite

The blue whale is the world's largest eating machine, growing up to 100 ft (30 m) long. Instead of teeth, it has filters made of baleen that hang from its upper jaw. When a blue whale finds a swarm of krill, it plows through them with its mouth open, while its throat expands like a balloon. It then closes its mouth and presses with its throat and tongue. The water is squeezed out, but the krill are trapped by the baleen and then swallowed. With this filtering system, a blue whale can gulp down 1 ton of food in a single meal.



Fish filters

With their mouths held open, these striped mackerel are using their gills to breathe and to feed. As they swim, they trap small animals with special gill rakers that are like teeth in a tiny comb. Filter-feeding fish include herrings, anchovies, and sardines, but this way of feeding is also used by some giants of the fish world. The largest of them all, the whale shark, grows up to 46 ft (14 m) long. It cruises through tropical seas, scooping up crustaceans, small fish, and squid.

Muscular pleats tighten to force water out of mouth

Heads down

Flamingos are the only birds that eat entirely by filter-feeding. They live in salty lakes and lagoons and use their strangely shaped beaks to collect shrimp and microscopic plants. A flamingo's tongue works like a pump, pushing water through the edges of its beak, which is lined with fibrous plates that trap its food. Young flamingos have straight beaks and feed on a milky fluid produced by their parents.

Flamingo feeding by sweeping upside-down beak from side to side



Mysterious life

This strange object is a chain of salps. These filter-feeders are close relatives of sea squirts, but instead of spending their adult lives fixed in one place, they move through the sea. Their bodies are shaped like hollow tubes, with muscles to pump water and a sieve to collect food. A chain of salps can contain hundreds of animals and grow many yards long and thicker than a human leg. Eventually, this breaks up, and individual salps produce eggs that start new chains.